

Federated Partial Identification: Comparing Six Aggregation Strategies

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Code: <https://github.com/watilde/Harmonia>

Abstract

I built an open-source tool for federated partial identification and tested six aggregation strategies on synthetic data. **One claim: inverse-width weighting achieved 15.5% tighter bounds than conservative aggregation under heterogeneity (CV=6.3%), converging to equivalence under homogeneity (CV=0.14%).** All strategies maintained constant 150-byte communication (3,200× reduction vs centralized). Tested on 1,130-2,709,803 synthetic patients across three scales.

Note: This demonstrates computational feasibility using synthetic Synthea data. Real-world validation requires institutional collaboration.

1. What I Built

A TypeScript tool that: - Computes Manski bounds (partial identification) at each site - Aggregates bounds using six weighting strategies - Transmits only 150 bytes total (4 numbers per site)

Six strategies tested: 1. Inverse-width: $w_k = (1/W_k)/\sum_j(1/W_j)$ 2. Sample-size: $w_k = n_k/N$ 3. Square-root: $w_k = \sqrt{n_k}/\sum_j\sqrt{n_j}$ 4. Logarithmic: $w_k = \log n_k/\sum_j\log n_j$ 5. Power: $w_k = n_k^\alpha/\sum_j n_j^\alpha$ 6. Conservative: $[\min_k \mathcal{L}_k, \max_k \mathcal{U}_k]$

2. Test Results (Synthetic Data)

Table 1: Measured Performance Across Three Scales

Scale	Patients	Sites	Inverse-Width	Sample-Size	CV (Hetero-Conservative)	CV (Heterogeneity)
1k	1,130	3	0.3903	0.3912	0.4616	6.3%
100k	235,222	3	0.3997	0.3997	0.4014	0.39%
2.8m	2,709,803	3	0.4000	0.4000	0.4009	0.14%

Key finding: Inverse-width provided 15.5% improvement over conservative at 1k scale (heterogeneous), converging to <0.3% difference at 2.8m scale (homogeneous).

Computational performance (2.8m patients): - Processing time: 12 seconds (225k patients/second) - Memory: 2-3 GB per site - Communication: 150 bytes (constant across all scales)

Communication reduction: - 1k scale: 201 KB $\rightarrow$ 150 bytes (1,341 $\times$) - 100k scale: 41.9 MB $\rightarrow$ 150 bytes (279,130 $\times$) - 2.8m scale: 482 MB $\rightarrow$ 150 bytes (3,200,000 $\times$)

3. Why This Pattern?

Minimax characterization (via KKT conditions): Inverse-width minimizes worst-case error when sites have different precision. Under heterogeneity, this provides measurable gains. Under homogeneity, all strategies converge because site-level errors become similar.

Formula: To minimize $\max_k \{w_k \cdot \epsilon_k\}$ subject to $\sum_k w_k = 1$, KKT stationarity requires $w_k \propto 1/\epsilon_k$, where ϵ_k is site k 's error (half-width).

This extends classical inverse-variance weighting from point estimates to interval bounds.

4. Limitations

Synthetic data only: I used Synthea synthetic healthcare data. Real hospital data may show different heterogeneity patterns. I have no access to real clinical data.

Three sites: Real networks (FDA Sentinel, PCORnet) have 10+ sites. Scaling behavior beyond 3 sites is untested.

Binary outcomes: Continuous outcomes and time-varying treatments require extensions.

No finite-sample inference: Bootstrap confidence intervals for federated bounds are future work.

Purpose: This is a technical demonstration, not clinical validation. Real-world applicability requires institutional collaboration.

5. Reproducibility

```
# Clone repository  
git clone https://github.com/watilde/Harmonia  
cd Harmonia
```

```
# Install dependencies  
npm install
```

```
# Build packages  
npm run build
```

```
# Run 1k scale test  
npm run test:partial-id:1k
```

```
# Run 100k scale test  
npm run test:partial-id:100k
```

```
# Run 2.8m scale test  
npm run test:partial-id:2.8m
```

Results saved to `research/modules/1-federated-partial-identification/experiments/results/`

6. What This Demonstrates

Computational feasibility: 2.8m patients in 12 seconds

Communication efficiency: 150 bytes (constant)

Strategy comparison: Six methods tested empirically

Heterogeneity effect: 15.5% improvement under $CV=6.3\%$

Real-world performance: Unknown (synthetic data only)

Clinical validity: No institutional validation

Large networks: 3 sites only

7. Related Work

Federated causal inference: Li et al. (2022) FACE, Zhang et al. (2023) FLAME, Xiong et al. (2023) FedCI - all require unconfoundedness or instrumental variables. Our partial identification maintains validity under arbitrary unmeasured confounding.

Meta-analysis: Fisher (1925), DerSimonian & Laird (1986) - inverse-variance weighting for point estimates. We extend to interval bounds via inverse-width weighting.

Partial identification: Manski (1990, 2003, 2007) - theory for single-site bounds. We provide federated aggregation framework.

8. Seeking Collaboration

I am an independent OSS engineer without access to real clinical data. This tool is ready for testing on actual multi-site hospital data. If you have OMOP-formatted data and IRB approval, I welcome collaboration to validate these findings in real-world settings.

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References

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 3. Li, S., et al. (2022). Federated causal inference in heterogeneous observational data. *arXiv:2202.12367*.
 4. Zhang, Y., et al. (2023). Privacy-preserving federated causal inference. *AAAI*, 37(12), 14589-14597.
 5. McMahan, B., et al. (2017). Communication-efficient learning from decentralized data. *AISTATS*.
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Data: Synthea synthetic healthcare data

Ethics: No IRB required (synthetic data only)